

now briefly discuss four distinct types of regulations that have been discussed in this context: (1) regulations that force platforms to reveal articles’ *provenance*; (2) *censorship* or tagging of misinformation; (3) *performance targets* that require the platform to keep misinformation below a given threshold; (4) *network regulations*, restricting the extent of ideological homophily or segregation introduced by platform algorithms intended to maximize engagement.

We consider the effects of these policies both on a fixed sharing network and when the platform can optimally choose the sharing network given public policy. For simplicity, we suppose the regulator’s objective is to decrease the virality of articles containing misinformation on the platform.²⁰ We will say the policy has an *ambiguous* effect when there exists a sharing network (and set of conditions) where the virality of misinformation *increases* following the regulatory policy.

6.1 Provenance

We consider a policy that requires the platform to reveal the original context or *provenance* of a piece of content. For example, provenance may point the user to a peer-reviewed medical study or the full discourse from which a quote was pulled. Such a policy allows users to verify (or “fact-check”) social media content easily and quickly.

We model a provenance policy by allowing some users to fact-check the article before making their sharing and dislike decisions. In particular, we assume revealing provenance allows each agent to check the veracity of the content instantly with probability $\rho > 0$, with higher values of ρ indicating more effective provenance policies.²¹ For simplicity, we assume fact-checks perfectly reveal whether the article has veracity $\nu = \mathcal{T}$ or $\nu = \mathcal{M}$. The user can then make a more informed (but still strategic) decision about whether to share the article or take one of the other two actions.

Proposition 5. *There exists $\rho^* < 1$ such that:*

- (a) *If $\rho > \rho^*$, for any fixed sharing network, the provenance policy reduces the virality of misinformation.*
- (b) *If $\rho < \rho^*$, the provenance policy has an ambiguous effect on the virality of misinformation.*
- (c) *If $\rho > \rho^*$, the provenance policy reduces the virality of misinformation even when the platform optimally chooses the sharing network conditional on the policy.*

We see that an effective provenance policy, which helps users identify misinformation with high likelihood, successfully reduces the circulation of this content. The reason is that when it is common knowledge that a piece of content contains misinformation, incentives toward sharing are uniformly diminished across all users on the platform. Strategic complementarities then further reduce sharing. However, less effective provenance policies ($\rho < \rho^*$) are *not* always beneficial and can sometimes backfire (part (b)), as illustrated in the next example.

²⁰Formally, the regulator minimizes the virality of content conditional on the article containing misinformation; in mathematical notation, the regulator minimizes $\mathbb{E}_{\max i^*} [\mathbf{S}_{i^*} \mid \nu = \mathcal{M}]$.

²¹In practice, certain demographic groups, such as users over 65 years old, appear more likely to accept misinformation without fact-checking, perhaps because of poor media interpretation skills (see Grinberg et al. (2019) and Guess et al. (2019)). Thus, provenance policies which provide a less clear pathway to fact-checking may lead certain social media users to fact-check and others to not.